

```
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```
import 'package:flutter/material.dart';

void main() {
  runApp(GalleryApp());
}

// Artwork class to easily represent the data we want to display
class Artwork {
  final String artist;
  final int year;
  final String imageAsset;
  final String title;

  const Artwork(this.artist, this.year, this.imageAsset, this.title);
}

// Storing artwork objects in a list to iterate through with buttons
final List<Artwork> _art_pieces = [
  Artwork('Unknown', 2021, 'assets/fence_rat.jpg', 'Rat Upon a Fence'),
  Artwork('Unknown', 2023, 'assets/RatGamer.jpg', 'Gamer Rat'),
  Artwork('Unknown', 2020, 'assets/him.jpg', 'Rat Upon a Couch'),
  Artwork(
    'Christos',
    2026,
    'assets/Smunga_Staring_at_Camera.jpeg',
    'Mr. Smunga Eyeing You',
  ),
  Artwork('Christos', 2021, 'assets/Bella.jpg', 'Bella Looking Cute'),
  Artwork('Britney', 2026, 'assets/Gub_in_box.jpeg', 'Mr. Gubbins in his box'),
  Artwork(
    'Unknown',
    2019,
    'assets/MJK.png',
    "Maynard James Keenan Messily Eating a Donut",
  ),
];

// Our primary class, stateless
class GalleryApp extends StatelessWidget {
  const GalleryApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(title: 'B + C Gallery', home: GalleryScreen());
  }
}

// Stateful widget representing the image and description
class GalleryScreen extends StatefulWidget {
  @override
  _GalleryScreenState createState() => _GalleryScreenState();
}

// State implementation
class _GalleryScreenState extends State<GalleryScreen> {
  int _currentAsset = 0; // for keeping track of which image should show
```

```
/// HELPER FUNCTIONS
```

```
void _incrementGallery() {  
  if (_currentAsset == _art_pieces.length - 1) {  
    setState(() {  
      _currentAsset = 0;  
    });  
  } else {  
    setState(() {  
      _currentAsset++;  
    });  
  }  
}
```

```
void _decrementGallery() {  
  if (_currentAsset == 0) {  
    setState(() {  
      _currentAsset = _art_pieces.length - 1;  
    });  
  } else {  
    setState(() {  
      _currentAsset--;  
    });  
  }  
}
```

```
// GUI design and implementation
```

```
@override
```

```
Widget build(BuildContext context) {
```

```
  return Scaffold(  
    appBar: AppBar(  
      title: const Text('Britney and Christos\' Photo Gallery'),  
      backgroundColor: Colors.deepPurple,  
      foregroundColor: Colors.white,  
    ),
```

```
    // Root of our scaffold, Centered Column
```

```
    body: Center(  
      child: Column(  
        children: [  
          // Begin by padding all sides  
          Padding(padding: EdgeInsets.symmetric(vertical: 10)),  
  
          // Next, display and expand the image  
          Expanded(  
            child: Card(  
              elevation: 14,  
              child: Padding(  
                padding: EdgeInsets.symmetric(horizontal: 50),  
                child: Image.asset(  
                  _art_pieces[_currentAsset].imageAsset,  
                  width: 300,  
                ),  
              ),  
            ),  
          ),  
        ],  
      ),
```

```
    // Padding between Image and Card
```

```
    Padding(padding: EdgeInsets.symmetric(vertical: 10)),
```

```
// Card for our image descriptions
Card.filled(
  child: Padding(
    padding: EdgeInsets.all(16),
    child: SizedBox(
      // Finding width of screen to properly size title card
      width: MediaQuery.sizeOf(context).width - 80,
      child: Column(
        crossAxisAlignment: CrossAxisAlignment.start,
        children: [
          Text(
            "${_art_pieces[_currentAsset].title}",
            style: TextStyle(
              fontSize: 28,
              fontWeight: FontWeight.bold,
            ),
          ),
          Text(
            "\n${_art_pieces[_currentAsset].artist} (${_art_pieces[_currentAsset].year})",
            style: TextStyle(fontSize: 18),
          ),
        ],
      ),
    ),
  ),
),
),
),
),

// Padding between card and buttons
Padding(padding: EdgeInsets.symmetric(vertical: 10)),

// Buttons, allow us to change the state of the above widgets
Row(
  mainAxisAlignment: MainAxisAlignment.spaceEvenly,
  children: [
    ElevatedButton(
      onPressed: _decrementGallery,
      child: const Text('<- Previous'),
      style: ElevatedButton.styleFrom(
        minimumSize: Size(125, 40),
        backgroundColor: Colors.blueGrey,
        foregroundColor: Colors.white,
      ),
    ),
    ElevatedButton(
      onPressed: _incrementGallery,
      child: const Text('Next ->'),
      style: ElevatedButton.styleFrom(
        minimumSize: Size(125, 40),
        backgroundColor: Colors.blueGrey,
        foregroundColor: Colors.white,
      ),
    ),
  ],
),

// Bottom padding to keep us away from the home bar / gesture region
Padding(padding: EdgeInsets.only(bottom: 80)),
],
```

```
    ),  
    ),  
  );  
}  
}
```